Body Performance Analytics and Intelligent Classification System

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1. Data Preparation and Exploratory Data Analysis (EDA).

1.1. Introduction

This report documents Part 1 of the Body Performance Analytics project. The objective of this phase is to understand the structure of the dataset, identify and resolve data quality issues, and perform a comprehensive Exploratory Data Analysis (EDA) to extract meaningful insights before proceeding to machine learning modeling in Part 2.

The analysis was conducted using Python with the following libraries:

- **pandas** — for data loading and manipulation
- **numpy** — for numerical computations
- **matplotlib and seaborn** — for data visualization
- **scikit-learn** — for preprocessing (StandardScaler)

1.2. Dataset Overview (Section 5.1)

The dataset used in this project is the Body Performance Dataset, sourced from Kaggle (dataset by kukuroo3). It contains physical fitness measurements collected from individuals who underwent standardized performance evaluations.

Upon loading the dataset, the following structural information was obtained:

- **Number of rows:** 13,393
- **Number of columns:** 12
- **Numeric columns (float64):** 10 — age, height_cm, weight_kg, body fat_%, diastolic, systolic, gripForce, sit and bend forward_cm, sit-ups counts, broad jump_cm
- **Categorical columns (object):** 2 — gender, class

Each row represents one participant's complete set of physical measurements and their corresponding performance classification. The dataset is well-structured with no immediately visible formatting issues upon initial inspection.

1.3. Column Understanding (Section 5.2)

Each column was examined individually to understand its meaning, expected data type, and validation constraints:

age — The participant's age in years. Numeric (integer, stored as float64). Must be positive and realistically between 15 and 80 years.

gender — The biological sex of the participant. Categorical. Accepts only two values: "M" for Male and "F" for Female.

height_cm — The participant's height measured in centimeters. Numeric (float). Expected between 100 and 250 cm.

weight_kg — The participant's body weight in kilograms. Numeric (float). Must be positive, typically between 30 and 200 kg.

body fat_% — The percentage of total body weight consisting of fat tissue. Numeric (float).

Must be between 0% and 60%; values outside this range are physiologically impossible.

diastolic — The diastolic blood pressure reading (lower number), measured in mmHg. Numeric. Expected between 40 and 120 mmHg.

systolic — The systolic blood pressure reading (upper number), measured in mmHg. Numeric. Must always be greater than the diastolic value, expected between 80 and 200 mmHg.

gripForce — Hand grip strength in kilograms, measured using a dynamometer. Numeric (float). Must be a non-negative value.

sit and bend forward_cm — A flexibility test measuring how far the participant can reach forward from a sitting position, in centimeters. Numeric (float). Can be negative if the participant cannot reach their toes — this is a valid physiological result, not an error.

sit-ups counts — Total number of sit-ups a participant can perform. Numeric (integer). Must be non-negative, typically between 0 and 80.

broad jump_cm — Distance achieved in a standing broad jump, measured in centimeters. Numeric (float). Must be positive, typically between 50 and 300 cm.

class — The overall physical performance category assigned to the participant. Categorical. Accepts four values: A (best), B, C, and D (lowest). This is the target variable for the classification task in Part 2.

1.4. Data Type Verification (Section 5.3)

The data types of all columns were verified using df.dtypes. The results confirmed that all 10 numeric columns are correctly stored as float64, and the 2 categorical columns (gender and class) are stored as object (string type).

Observation: The age column is stored as float64 instead of integer, which is technically acceptable since no decimal precision is lost in the analysis. The sit-ups counts column is similarly stored as float64 despite being a whole-number count — this does not affect correctness. No type conversion was required as the data behaves correctly for all downstream operations.

1.5. Missing Values Analysis (Section 5.4)

A complete missing value analysis was performed by computing both the count and percentage of missing entries for every column.

Result: The dataset contains zero missing values across all 12 columns. Every cell in all 13,393 rows is fully populated.

Although no imputation was needed in this dataset, the following strategy was defined for future reference:

- **For numeric columns:** replace missing values with the median (more robust than the mean in the presence of outliers)
- **For categorical columns:** replace missing values with the label "Unknown"
- **For rows where more than 50% of values are missing:** drop the entire row

1.6. Duplicate Detection (Section 5.5)

The dataset was scanned for exact duplicate rows using df.duplicated().

Result: 1 duplicate row was detected. This row was identical to another existing record in every column, suggesting it was erroneously entered twice. The duplicate was removed using df.drop_duplicates(), reducing the dataset from 13,393 to 13,392 rows.

1.7. Data Validity Checks (Section 5.6)

Beyond missing values and duplicates, the data was inspected for logically impossible or physiologically unrealistic values:

Check	Condition Tested	Violations Found
Age	< 0 or > 100 years	0 rows
Weight	<= 0 kg	0 rows
Height	<= 0 cm	0 rows
Body fat %	< 0% or > 60%	1 row
Blood pressure	systolic <= diastolic	5 rows

Table 1

Body fat violation: One record had a body fat percentage outside the physiologically valid range (0-60%). This value is biologically impossible and was removed.

Blood pressure violation: Five records had systolic blood pressure readings that were less than or equal to the diastolic reading. This is physiologically impossible — systolic pressure is always higher than diastolic pressure in any living person. These 5 rows were removed.

Action taken: After removing the duplicate and all invalid records, the dataset was reduced to 13,386 clean rows — a loss of only 0.05% of the original data, which is entirely acceptable.

1.8. Univariate Analysis (Section 5.7)

Each numeric column was analyzed individually using descriptive statistics. The full summary is presented in the table below:

Feature	Mean	Median	Std Dev	Min	Max
age	36.78	32.0	13.63	21.0	64.0
height_cm	168.56	169.2	8.43	125.0	193.8
weight_kg	67.45	67.4	11.95	26.3	138.1
body fat_%	23.24	22.8	7.26	3.0	54.9
diastolic	78.80	79.0	10.74	6.0	126.0
systolic	130.27	130.0	14.71	77.0	201.0

Feature	Mean	Median	Std Dev	Min	Max
gripForce	36.96	37.9	10.62	0.0	70.5
sit and bend forward_cm	15.21	16.2	8.46	-25.0	213.0
sit-ups counts	39.77	41.0	14.28	0.0	80.0
broad jump_cm	190.13	193.0	39.87	0.0	303.0

Table 2

Interpretation of Each Feature

Age: The mean age is approximately 37 years with a standard deviation of 13.6. The median (32) is lower than the mean (36.8), indicating a slight right skew — there are more younger participants but a tail of older individuals pulling the average upward. The range is 21-64 years.

Height: The mean height is 168.6 cm, consistent with a mixed-gender population. The small standard deviation (8.4 cm) and nearly equal mean and median indicate an approximately symmetric distribution. The wide range (125-193.8 cm) reflects natural human variation.

Weight: The mean weight of 67.45 kg is almost identical to the median (67.4 kg), suggesting a very symmetric distribution. The standard deviation of 11.95 kg reflects moderate variation.

Body Fat %: A mean of 23.24% with a right skew (skewness = 0.36) — most participants have moderate body fat, but some individuals with very high body fat percentages pull the mean slightly upward.

Diastolic Blood Pressure: Mean of 78.8 mmHg, within the normal clinical range. Very slight negative skew (-0.16), indicating a mostly symmetric distribution.

Systolic Blood Pressure: Mean of 130.3 mmHg, aligning with prehypertension clinical boundaries. Nearly symmetric (skewness = -0.05). Standard deviation of 14.7 mmHg shows moderate spread.

Grip Force: Mean of 36.96 kg, almost identical to the median (37.9 kg). Extremely low skewness (0.018) indicates a nearly perfect symmetric distribution.

Sit and Bend Forward: This feature has the most notable range: -25 cm to 213 cm. The negative values are valid — they indicate participants who cannot reach their toes. The positive skewness (0.786) indicates a right-skewed distribution.

Sit-ups Count: Mean of 39.77, median of 41. Negative skewness (-0.47) means most participants cluster around 30-50 sit-ups with fewer achieving very low counts.

Broad Jump: Mean of 190.13 cm, median of 193 cm. Negative skewness (-0.42) and a wide standard deviation of ~40 cm. The minimum of 0 cm indicates at least one extreme outlier.

1.9. Distribution Analysis — Histograms (Section 5.8)

A grid of 10 histograms (4 rows x 3 columns) was generated, one for each numeric column, using 30 equal-width bins. Each histogram shows how frequently values fall within each interval.

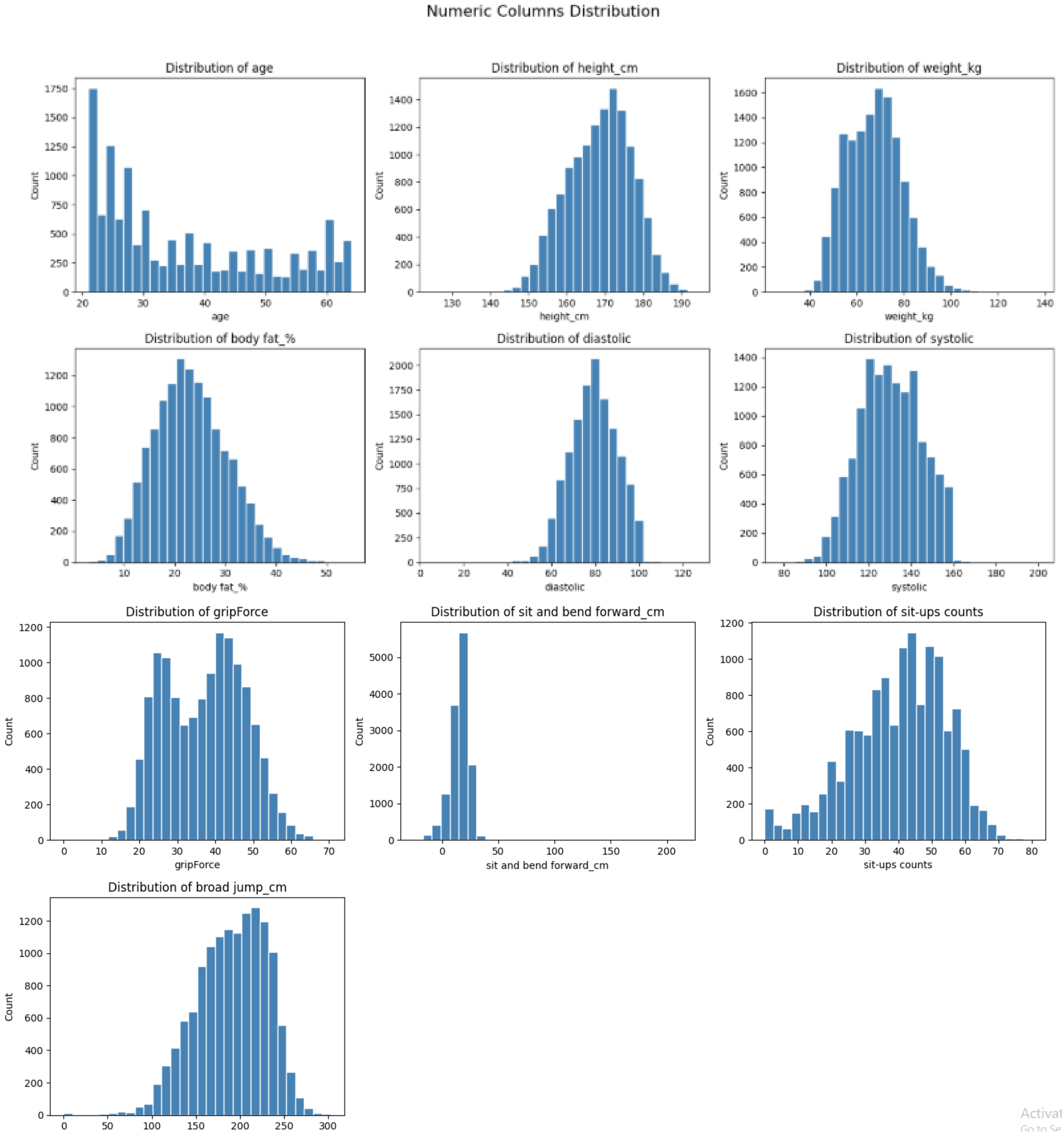

Figure 1

Interpretation of Each Histogram

Age histogram: Shows a right-skewed distribution. The highest frequency bars are concentrated in the 20-35 age range, with a long tail extending to 64 years. This confirms that younger participants dominate the dataset.

Height_cm histogram: Displays a roughly bimodal distribution with two overlapping peaks

— one around 160 cm (likely female participants) and one around 175 cm (likely male participants). This bimodality is expected in a mixed-gender dataset.

Weight_kg histogram: Approximately normal and symmetric, centered around 65-70 kg. Slightly right-skewed with a long tail toward heavier weights.

Body fat_% histogram: Right-skewed distribution. The majority of participants fall in the 15-30% range, with declining frequency toward higher values.

Diastolic histogram: Approximately symmetric and bell-shaped, centered around 75-80 mmHg. A few extreme low values are visible in the left tail.

Systolic histogram: Approximately symmetric and normal-looking, centered around 125-135 mmHg. The distribution is slightly wider than diastolic, reflecting greater natural variation.

GripForce histogram: Symmetric and bell-shaped, closely resembling a normal distribution. Center is around 35-40 kg.

Sit and bend forward_cm histogram: Right-skewed with the peak around 10-20 cm. Has a long right tail and a noticeable left extension into negative values, representing participants with poor flexibility.

Sit-ups counts histogram: Approximately symmetric with a slight left skew. Most values are concentrated in the 25-55 range.

Broad jump_cm histogram: Slightly left-skewed and approximately normal, centered around 180-210 cm.

1.10. Categorical Frequency Plots (Section 5.8 — Categorical)

Two bar charts were generated to examine the two categorical variables in the dataset.

Gender Distribution Bar Chart: The chart shows the count of Male (M) and Female (F) participants. The dataset contains a moderate imbalance, with more male participants than female. This gender imbalance is important to note because many physical fitness metrics — such as grip force and broad jump — differ significantly between genders, and gender should be included as a feature in the models.

Performance Class Distribution Bar Chart: The four performance classes (A, B, C, D) are displayed in order from best to worst. The chart reveals that the classes are relatively balanced across all four categories, which is highly favorable for machine learning classification tasks. A heavily imbalanced class distribution would require special handling techniques such as oversampling, but this dataset does not require such intervention.

1.11. Scatter Plots (Section 5.8 — Relationships)

Four scatter plots were created to visualize pairwise relationships between important features:

Age vs Body Fat %: The scatter plot shows a positive trend — as age increases, body fat percentage tends to increase. The relationship is moderate and expected physiologically, as metabolic rate decreases with age. The wide scatter indicates that age alone is not a perfect predictor of body fat.

Grip Force vs Broad Jump: This plot reveals a strong positive correlation. Participants with higher grip force tend to achieve longer broad jumps. Both metrics reflect overall muscular strength, which explains this relationship. The data points form a clear upward-trending cloud.

Sit-ups Count vs Broad Jump: Another positive relationship is visible here. Participants who can perform more sit-ups also tend to jump farther. Both activities require core and lower body strength, creating this natural correlation.

Height vs Weight: A clear positive correlation exists, as expected. Taller individuals tend to weigh more. Two sub-clusters may be visible (male vs female subgroups), which reinforces the bimodal nature observed in the height histogram.

1.12. Outlier Detection — Boxplots (Section 5.9)

A grid of 10 boxplots was generated, one per numeric column. The boxplot is a powerful visualization for detecting outliers.

How to Read a Boxplot

- The central box spans the Interquartile Range (IQR) — the middle 50% of the data
- The horizontal line inside the box is the median
- The whiskers extend to 1.5x IQR beyond the box edges
- Any data points beyond the whiskers are plotted as individual dots and considered outliers

Observations from the Boxplots

Age: A few outlier dots appear at the upper end (above ~60 years), but these are physiologically realistic values.

Height: Very few outliers; the distribution is tight and clean.

Weight: A noticeable number of upper outliers representing very heavy participants. These are realistic but extreme.

Body fat_%: Upper outliers present, representing individuals with very high body fat percentages.

Diastolic: Outliers on both ends — very low readings (near 0) and very high readings. Extremely low values may be measurement errors.

Systolic: Upper outliers visible, with some readings exceeding 180 mmHg.

GripForce: Both lower outliers (very weak individuals) and upper outliers (exceptionally strong participants).

Sit and bend forward_cm: Significant upper outliers — one value reaching 213 cm is almost certainly a data entry error.

Sit-ups counts: A few outliers at both extremes.

Broad jump_cm: Lower outliers visible near 0 cm, very likely measurement errors.

Decision: IQR Capping Method

Rather than removing outlier rows (which would reduce the dataset size), the IQR Capping Method was applied. Any value below Q1 - 1.5xIQR was set to that lower bound, and any value above Q3 + 1.5xIQR was set to that upper bound. This approach preserves all 13,386 rows while preventing extreme values from distorting model training.

1.13. Correlation Analysis — Heatmap (Section 5.10)

A correlation heatmap was generated using numeric_data.corr() with the coolwarm color palette. Each cell shows the Pearson correlation coefficient between two features, ranging from -1 to +1. Red cells indicate strong positive correlations, blue cells indicate negative correlations, and neutral cells indicate no relationship.

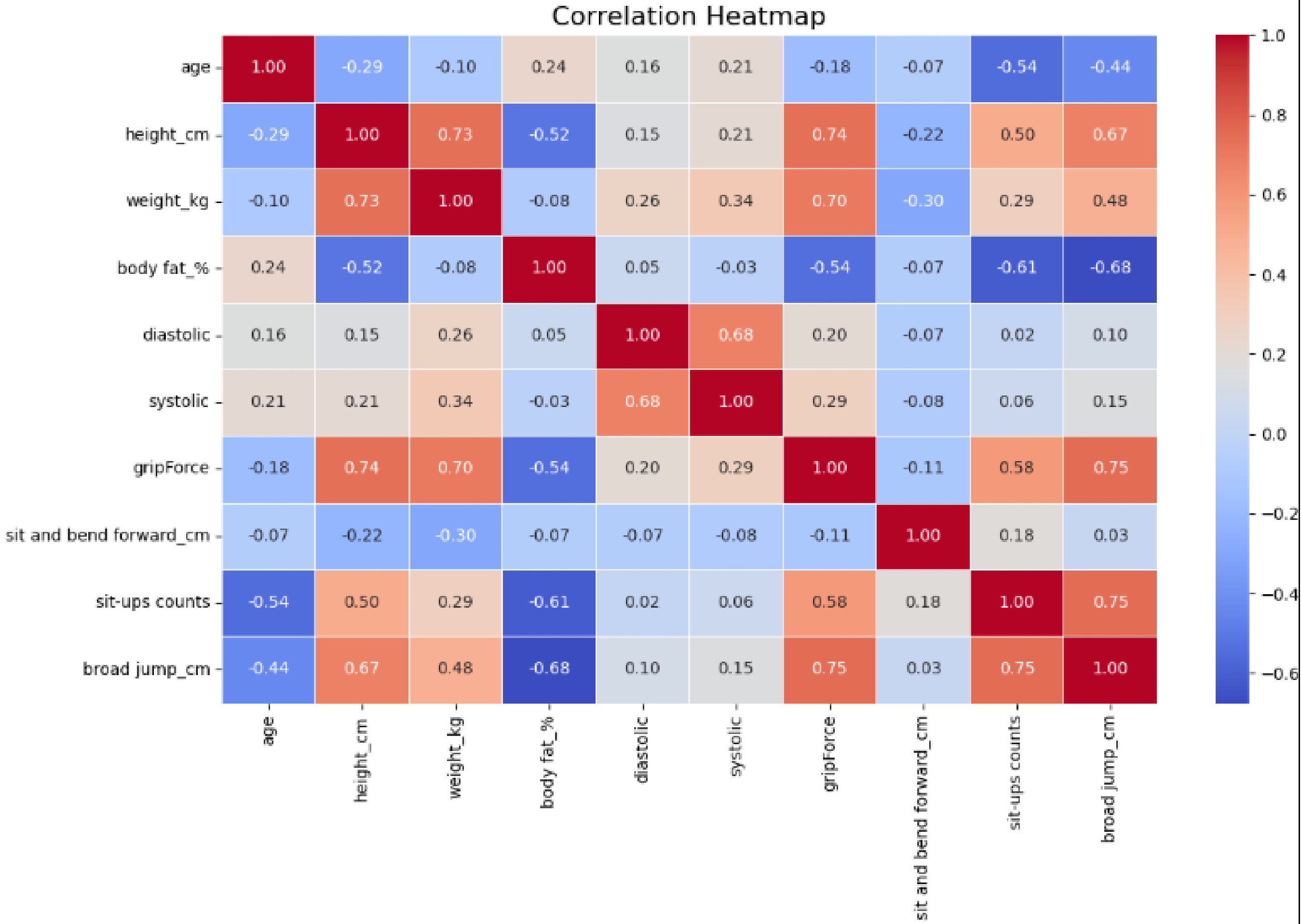

Figure 2

Top 5 Strongest Correlations

Feature Pair	Pearson Correlation
sit-ups counts <-> broad jump_cm	0.75
broad jump_cm <-> gripForce	0.75
height_cm <-> gripForce	0.74
weight_kg <-> height_cm	0.73
gripForce <-> weight_kg	0.70

Table 3

Interpretation

The strongest correlation (0.748) exists between sit-ups count and broad jump distance, confirming that muscular endurance and explosive power are closely related fitness dimensions. The second strongest (0.747) is between broad jump and grip force, highlighting that participants with overall higher physical fitness excel across multiple domains simultaneously.

Height and grip force show a strong correlation (0.735) because taller individuals typically have larger muscle mass. Similarly, height and weight correlate strongly (0.735) for the same physiological reason.

An important implication of these strong correlations is multicollinearity — when building linear models in Part 2, highly correlated features may inflate coefficient variance and reduce model interpretability. Feature selection strategies should be considered.

Notably, blood pressure (diastolic, systolic) shows relatively weak correlations with performance features, suggesting that blood pressure alone is not a strong predictor of physical fitness class.

1.14. Data Preprocessing Steps Applied

After completing EDA, the following preprocessing steps were applied to prepare the dataset for Part 2 modeling:

Step	Action	Status
1 — Outlier Capping	IQR capping on all 10 numeric columns	Done
2 — Encode gender	Male = 1, Female = 0	Done
3 — Encode class	A = 3, B = 2, C = 1, D = 0	Done
4 — Feature Scaling	StandardScaler applied to all feature columns	Done
5 — Data Type Verification	All columns verified as numeric post-encoding	Done

Table 4

Encoding rationale: The class column was encoded using an ordinal scheme (A=3, B=2, C=1, D=0) to preserve the natural performance ranking inherent in the class variable.

Scaling rationale: StandardScaler transforms each feature to have a mean of 0 and standard deviation of 1. This is essential for distance-based algorithms like K-Nearest Neighbors and Support Vector Machines, which are sensitive to differences in feature scale.

1.15. Final EDA Summary (Section 5.11)

Five Key Insights Discovered

1. Grip force and broad jump have the strongest feature relationship in the dataset (r = 0.747). This confirms that overall physical strength manifests consistently across multiple performance measurements.
2. Body fat percentage increases with age, as demonstrated by the scatter plot. This trend is physiologically expected, as metabolic rate decreases with age.
3. Males generally exhibit higher grip force values than females, which creates the bimodal shape observed in several histograms and contributes to the strong height-grip force correlation.
4. Sit-ups count is one of the strongest predictors of performance class, as muscular endurance is directly reflected in the class scoring system.

5. The performance class distribution is approximately balanced across A, B, C, and D, making this dataset well-suited for standard classification algorithms without requiring class-balancing techniques.

Five Potential Data Quality Problems

6. Outliers in sit and bend forward_cm — one value reaching 213 cm is almost certainly a data entry or recording error.
7. Outliers in systolic blood pressure — readings exceeding 200 mmHg represent dangerously high hypertension and may be measurement errors.
8. One body fat record exceeded 60% — this is physiologically impossible and was removed from the dataset.
9. Five blood pressure records had systolic <= diastolic — this is medically impossible and represents a measurement or recording failure.
10. The minimum broad jump value of 0 cm suggests that at least one participant either could not perform the test or a recording error occurred.

Recommended Preprocessing Steps

Step	Action	Status
Remove invalid body fat record	Drop row with body fat% > 60	Done
Remove invalid blood pressure records	Drop rows where systolic <= diastolic	Done
Remove duplicate rows	Drop exact duplicate entries	Done
Cap outliers	IQR capping for all numeric columns	Done
Encode categorical variables	Gender -> 0/1, Class -> 0/1/2/3	Done
Standardize numeric features	StandardScaler (mean=0, std=1)	Done

Table 5

1.16. Final Dataset State

After completing all data preparation steps, the dataset is in the following state:

Property	Value
Total rows	13,386 (removed 7 rows: 1 duplicate + 6 invalid)
Total columns	12
Missing values	0
Duplicate rows	0
Data types	All columns are numeric (float64 or int64)
Outliers	Capped using IQR method
Encoding	Gender and class converted to numeric labels
Scaling	All feature columns normalized via StandardScaler

Table 6

The dataset is fully clean, validated, and prepared for machine learning model training in Part 2.

2. Transition from EDA to Machine Learning.

2.1. After EDA – Preparing for Machine Learning (Imports Section)

After completing the exploratory data analysis, the next step was to prepare the machine learning environment. At this stage, the goal was not to train models immediately, but to carefully select the appropriate tools required for data **preprocessing, model training, validation, and evaluation**. This step is similar to preparing the tools before starting an experiment, where each tool has a specific role in ensuring that the experiment is conducted correctly and the results are reliable.

The sklearn.model_selection module was used for splitting the dataset into training and testing sets and performing cross-validation to ensure reliable model evaluation. This helps prevent overfitting and ensures that the model performs well on unseen data.

The sklearn.preprocessing module was used for feature scaling and encoding categorical variables. Feature scaling using MinMaxScaler was particularly important because some machine learning algorithms such as KNN and SVM are distance-based algorithms and require normalized data.

Different machine learning algorithms were imported, including K-Nearest Neighbors, Decision Trees, Random Forest, Support Vector Machines, Neural Networks, and Linear Regression. Multiple models were used in order to compare different learning approaches and determine which algorithm performs best on the dataset.

Finally, evaluation metrics such as accuracy, precision, recall, F1-score, mean squared error, and R-squared were imported to measure model performance for both classification and regression tasks.

Overall, this step prepared the environment and tools required for building and evaluating the machine learning models.

2.2. Data Preparation for Machine Learning

After setting up the machine learning environment, the dataset was prepared for model training. A copy of the dataset was created to avoid modifying the original data.

Feature engineering was performed to improve model performance and reduce redundancy in the dataset. Weight and height were replaced with Body Mass Index (BMI), which is a more meaningful indicator of body composition. Additional features were created such as strength index, flexibility score, and cardio index to capture relationships between physical fitness measurements.

These engineered features help the machine learning models understand interactions between variables rather than relying only on raw measurements.

After feature selection and feature engineering, the dataset contained 10 input features that were used for both classification and regression tasks.

2.3. Train-Test Split

A standard **80:20 split** was applied using stratification to preserve class distribution. Importantly, the split was performed **before any preprocessing**, ensuring that no information from the test set leaked into the training process.

2.4. Pipeline Design and Data Leakage Prevention

To maintain methodological rigor, a **Pipeline** was used to combine:

- MinMaxScaler
- The machine learning model

This approach ensures that scaling is applied **only on training data**, effectively preventing **data leakage** and improving the reliability of evaluation results.

2.5. Machine Learning Models – Classification

Several classification models were trained and compared in this project, including:

2.5.1. K-Nearest Neighbors (KNN)

After tuning the number of neighbors using cross-validation, the optimal value of **K = 24** was selected. The model achieved the following performance on the test dataset:

- **Accuracy:** 0.6718
- **Precision:** 0.6901
- **Recall:** 0.6718
- **F1 Score:** 0.6726

These results indicate that the KNN model achieved moderate performance. Since KNN is a distance-based algorithm, its performance depends heavily on the distribution of the data. The moderate accuracy suggests that the classes in the dataset overlap and are not clearly separable using simple distance-based classification.

The strength of the KNN model is that it is simple and does not assume any underlying distribution of the data. However, its weakness is that it does not perform well when the dataset contains complex or non-linear relationships.

2.5.2. Decision Tree Classifier

The Decision Tree model was tuned using different values of maximum depth, and the optimal depth was found to be max_depth = 11. The model achieved:

- **Accuracy:** 0.6833
- **Precision:** 0.6903
- **Recall:** 0.6833
- **F1 Score:** 0.6838

The Decision Tree performed slightly better than KNN. One major advantage of Decision Trees is that they provide feature importance, which helps in understanding which variables are most influential in the classification process.

The feature importance results showed that the most important features were:

1. Sit and bend forward (flexibility)
2. Sit-ups counts
3. Strength index
4. Body fat percentage
5. Age
6. BMI

This indicates that flexibility and strength-related features play the most significant role in determining fitness classification. The strength of Decision Trees is that they are easy to interpret and can model non-linear relationships. However, they can suffer from overfitting if the tree depth is too large.

2.5.3. Random Forest Classifier

The Random Forest model achieved:

- **Accuracy:** 0.7341
- **F1 Score:** 0.7332

This was significantly higher than both KNN and Decision Tree models.

Random Forest improves performance by combining multiple decision trees and averaging their results. This reduces overfitting and improves generalization. The higher accuracy indicates that ensemble methods are more suitable for this dataset.

The strength of Random Forest is that it handles non-linear relationships well and is more robust than a single decision tree. One disadvantage is that it is less interpretable compared to a single decision tree.

2.5.4. Support Vector Machine (SVM)

Different kernels were tested for the SVM model:

Kernel	Accuracy	F1 Score
Linear	0.6139	0.6129
RBF	0.6957	0.6963
Polynomial	0.6998	0.7001

Table 7

The linear kernel produced the lowest accuracy, while the RBF and polynomial kernels performed significantly better. This indicates that the dataset is **non-linear**, and non-linear kernels are better suited for classification.

After tuning the regularization parameter **C**, performance improved further during cross-validation, indicating that hyperparameter tuning is important for SVM performance.

The strength of SVM is that it performs well in high-dimensional spaces and can model complex decision boundaries. However, it requires careful parameter tuning.

2.5.5. Neural Network Classifier

Different neural network architectures were tested:

Architecture	Accuracy	F1 Score
(50,)	0.6613	0.6595
(100,)	0.6815	0.6819
(50, 25)	0.7013	0.7042
(100, 50)	0.7106	0.7124

Table 8

The best architecture was **(100, 50)**, which achieved the highest performance among all classification models.

Cross-validation results showed that the Neural Network achieved approximately **74% accuracy**, making it the best overall classification model.

Neural networks performed best because they can model complex non-linear relationships between features, which are common in human physical performance data.

The strength of neural networks is their ability to learn complex patterns and interactions between variables. However, they require more training time and parameter tuning compared to simpler models.

2.6. Regression Models – Predicting Broad Jump Distance

2.6.1. Linear Regression

Linear Regression achieved the following results:

- **MSE:** 341.56
- **RMSE:** 18.48
- **R^2:** 0.7919

An R^2 value of **0.79** means that the model explains approximately **79% of the variance** in broad jump distance, which is considered a strong result.

The regression coefficients showed that the most important predictors were:

- Grip strength (positive effect)
- Sit-ups counts (positive effect)
- Body fat percentage (negative effect)
- Age (negative effect)
- Flexibility (positive effect)

This means that strength and flexibility increase performance, while body fat and age reduce performance.

2.6.2. Decision Tree Regression

Decision Tree Regression achieved:

- **MSE:** 448.13
- **RMSE:** 21.17
- **R^2:** 0.7269

This is slightly worse than Linear Regression, which suggests that the relationship between the features and the broad jump distance is relatively linear rather than highly non-linear.

2.7. Cross-Validation Results and Model Stability

To ensure that the model performance results are reliable and not dependent on a single train–test split, **K-Fold Cross Validation** was applied. In this method, the dataset was divided into **K equal folds (K = 5)**. The model was trained five times, each time using four folds for training and one fold for testing. The final model performance was calculated as the average accuracy across all folds.

This approach provides a more reliable evaluation because every data point is used for both training and testing, reducing the risk of overfitting and giving a better estimate of how the model will perform on unseen data.

The cross-validation results

Model	Cross-Validation Accuracy
KNN	0.6738
Decision Tree	0.6825
SVM	0.6951
Neural Network	0.7432

Table 9

The small difference between cross-validation accuracy and test accuracy indicates that the models are not overfitting and have good generalization performance.

3. Part 3: Performance Evaluation and Model Testing

3.1. Classification

By comparing the classification models, the Random Forest model achieved the best overall performance. It obtained the highest accuracy and F1 score among all models, indicating strong classification capability and balanced performance across all classes. Therefore, Random Forest was selected as the best classification model for this task.

Figure 3

The confusion matrix shows that most predictions are correctly classified along the diagonal, indicating good overall model performance. The model performs very well in predicting classes A and D, with a high number of correct classifications. However, there is noticeable confusion between classes C and D, where several instances are misclassified between these two categories. This suggests that these classes share similar characteristics, making them harder to distinguish. Overall, the Random Forest model demonstrates strong classification performance with some limitations in separating closely related classes.

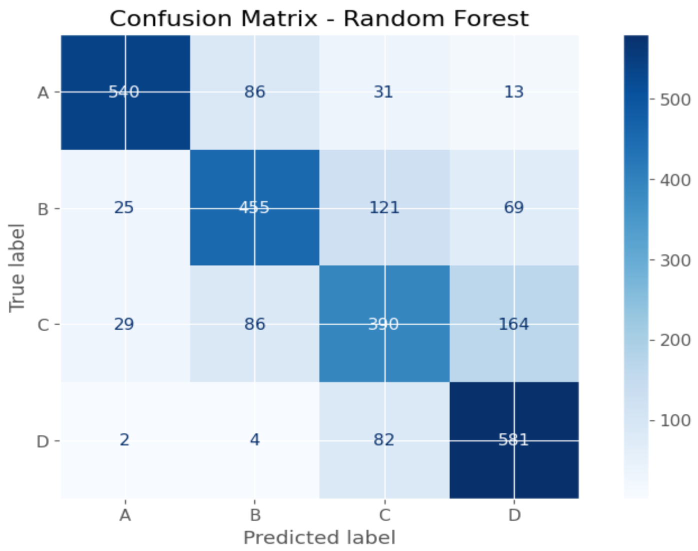

Figure 4

3.2. Regression

Regression Models Evaluation:

Model	MSE	RMSE	R2 Score
Linear Regression	341.5611	18.4814	0.7919
Decision Tree Regression	448.1381	21.1693	0.7269

Table 10

The results clearly indicate that Linear Regression outperforms the Decision Tree Regressor, as it produces lower prediction errors and a higher R^2 score, making it the most suitable model for regression in this study and the figure shows that

Figure 5

Best Regression Model:

Model	Linear Regression
MSE	341.5611
RMSE	0.7919
R2 Score	18.4814

Table 11

Conclusion

This project aimed to analyze the Body Performance dataset and build machine learning models to predict both performance classification and physical capability measures. A complete machine learning pipeline was implemented, including data preprocessing, exploratory data analysis (EDA), feature engineering, model training, and evaluation.

During the data preparation phase, several preprocessing techniques were applied, including handling data quality issues and creating new meaningful features such as BMI, strength index, and cardio index. The dataset was found to be balanced across all classes, which allowed the use of classification models directly without the need for class balancing techniques, ensuring that the models do not become biased toward any specific class.

In addition, outliers were detected and handled using clipping to reduce their impact on model performance. Furthermore, the exploratory data analysis (EDA) helped identify the most important features, understand data distributions, and uncover relationships between variables, which guided preprocessing steps and significantly improved model performance.

Multiple machine learning models were trained and evaluated for the classification task, including K-Nearest Neighbors, Decision Tree, Random Forest, Support Vector Machine, and Neural Network. The results showed that the Random Forest model achieved the best overall performance, with the highest accuracy and F1 score, indicating strong classification capability and balanced predictions across all classes.

The confusion matrix analysis confirmed that the model performs well in most classes, particularly classes A and D, while some confusion was observed between classes C and D due to their similarity.

For the regression task, Linear Regression achieved the best performance compared to other models, with a high R^2 score and relatively low prediction error, indicating a strong ability to predict the target variable.

Additionally, cross-validation experiments demonstrated that the models maintain consistent performance across different data splits, supporting their reliability and generalization ability.

In conclusion, this project demonstrates how machine learning techniques can be effectively applied to real-world datasets to extract insights and build accurate predictive models, highlighting the importance of proper data preprocessing, feature engineering, and model selection.

AI Disclosure: This report was produced with the assistance of AI tools, declared in accordance with the academic integrity policy stated in the Course Project Handbook.